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XGBoost and DecisionTree Algorithms for Fraud Detection

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ABSTRACT

As credit card usage continues to expand, so does the potential for fraudulent transactions. Fraudsters constantly evade capture due to the time-consuming and error-related mechanics behind manual fraud detection. Noticing this, we decided to employ artificial intelligence in Python to combat the problem. Throughout the study, we conducted a comparative analysis of XGBoost (XGB) and DecisionTree (DT) machine-learning algorithms, training three XGB and DT models each, and leveraging different performance metrics to evaluate their effectiveness in classifying transactions. Due to the imbalanced nature of our dataset, we also executed random over-sampling (ROS) and random under-sampling (RUS) techniques to generate equality between the classes, eliminating inherent bias. We approached the assessment of the machine learning models in 2 ways. We determined the best models once grounded in a standard approach, and once based on a novel approach: prioritize high recall and forgive low precision. Using metrics such as accuracy, precision, recall, f1-score, and confusion matrices permitted us to designate unmanipulated XGB as the greatest model based on the conventional evaluation strategy and RUS XGB from our original method.

Introduction

Credit card fraud refers to the unlawful use of another individual's credit card with the intent to remove funds or make purchases tied to their account [1]. In the year 2023, the Federal Trade Commission reported roughly 400,000 individuals filed for credit card identity theft [2]. Identity theft can lead to a deep trench of hopelessness and stress for victims. They may experience financial difficulties, hopelessness, isolation, and family discord [3]. The vast amount of stolen credit card information and severe physical and mental ramifications highlight the vitality of implementing new and efficient fraud detection methods. As the transition from cash to credit cards surges, the possibility for fraud to occur also escalates. From Q1 2020 - Q1 2023, the number of credit cards grew by nearly 63.6 million [4].

We decided to attack the issue and promote financial security with machine learning. Combing through each transaction with the assistance of artificial intelligence will elevate the rate of classification significantly compared to manual classification, saving hundreds of hours in the long term.

Background

We reviewed numerous approaches to the issue of fraud detection. Awoyemi et al. conducted a comparative analysis of machine learning algorithms Naive Bayes, K-nearest neighbors, and logistic regression [5]. They carried out a hybrid of under-sampling and over-sampling to formulate two sets of distribution (10:90 and 36:66) due to the extreme skew toward the negative class (legitimate transactions). After evaluating their models' using accuracy, sensitivity, specificity, precision, Matthew's correlation coefficient, and balanced classifi-

cation rate, they discovered executing hybrid sampling significantly augmented two of their classifiers' performances: Naive Bayes and K-nearest neighbors. Logistic regression, on the other hand, didn't display signs of enhancement, although it did outperform the other classification algorithms in the un-sampled distribution. The group found that, across all 3 dataset distributions, K-nearest neighbors performed the greatest, followed by Naive Bayes, and then logistic regression. Another approach is highlighted by the work of Behera and Panigrahi [6]. They introduced a novel technique fusing fuzzy clustering and neural networks in the department of credit card fraud detection. They observed the performance of the model to contain up to 93.90% true positives and less than 6.10% false positives. Choi and Lee compared the effectiveness between their machine learning model and artificial neural network [7]. The machine learning model consisted of clustering processes expected maximization (EM), simple, FarthestFirst, XMeans, and DensityBased algorithms and classification algorithms Naive Bayes, support vector machine (SVM), regression, OneR, C4.5, and RandomForest. They reported the machine learning method yielded an average detection rate of .98618, whereas the artificial neural network yielded an average detection rate of 0.77228. Another approach to the situation is explored by Fang et al. [8]. They proposed using a Light Gradient Boosting Machine (LightGBM) model as an alternative to the traditional RandomForest and Gradient Boosting Machine algorithms to combat the issue. LightGBM is rooted in the normal gradient boosted decision tree algorithm. It achieves enhanced accuracy by leveraging histogram algorithm and leaf-wise growth strategy. Fang et al.'s study demonstrated LightGBM results in the greatest AUC score at 99%, with RF at 98%, and Gradient Boosting Machine at 84%. Despite the visually miniscule difference, the variation can be the difference between hundreds of transactions being correctly classified.

A common trend among datasets concerned with credit card fraud detection is class imbalance. This occurrence is characterized by many more instances of some classes over others [9]. Countless methods can adjust the training sets to compensate for the bias caused by the class imbalance. The balancing processes tend to fall into two categories: under-sampling and over-sampling. Rendón et al. gauge the potency of multiple balancing techniques [10]. Their research concluded ROS as a competitive option against synthetic minority over-sampling technique (SMOTE) when applied on artificial neural networks. Additionally, hybrid options consisting of SMOTE didn't reveal statistically significant signs of improvement when compared to unmanipulated SMOTE.

Dataset

For the construction of our models, we employed Kaggle's *Credit Card Fraud Detection* dataset. The dataset consists of fraudulent and legitimate credit card transactions made by European cardholders over a 2-day period in September 2013 [11]. The content of the dataset is completely numerical.

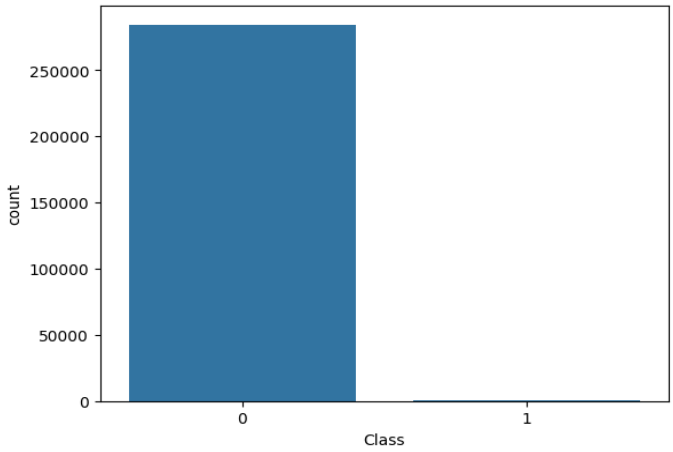


Figure 1. Number of Legitimate and Fraudulent Transactions

The dataset details a quantity of 284,807 transactions with a significantly imbalanced dynamic: 284,315 or ~99.82% are legitimate and 492 or ~.172% of the transactions are fraudulent. Figure 1 demonstrates the severe class imbalance in the dataset using a bar graph.

284,807 rows (each transaction) and 31 rows (Time, V1-V28, Amount, and Class) compose the dataset. The dataset was subject to principal component analysis to maintain the confidentiality of the cardholders [11].

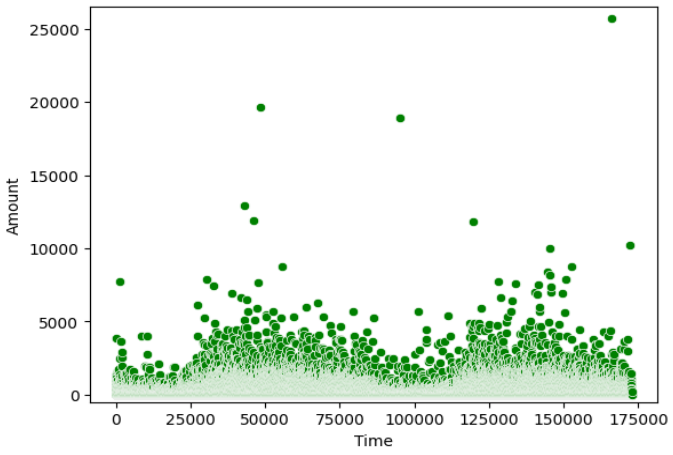


Figure 2. Amount vs Time for the Legitimate Class

The average amount for the legitimate transactions is ~$88.29 with a range of $25,691.16. Figure 2 demonstrates the distribution of legitimate transactions over the 2-day period the dataset was recorded on.

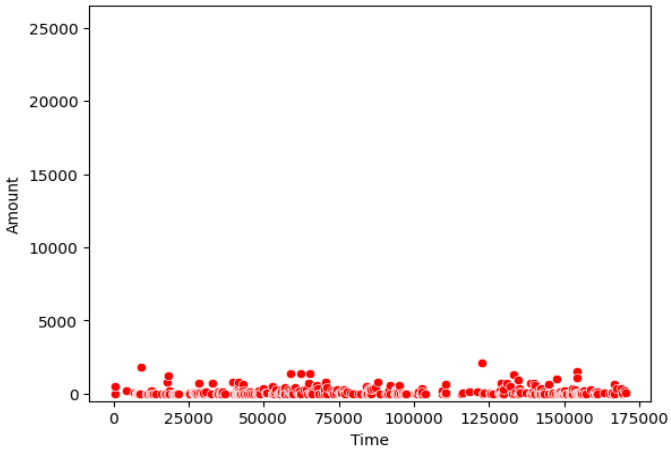


Figure 3. Amount vs Time for the Fraudulent Class

Conversely, the average amount for fraudulent transactions is ~$122.21 with a range of $2125.87. Figure 3 illustrates the distribution of fraudulent transactions during the two days.

When constructing the DT and XGB models, we split the dataset into 80% training data and 20% testing data, using all the features provided within the dataset.

Methodology and Models

We began by constructing baseline DT and XGB models.

A DT functions as a supervised machine-learning algorithm with classification and regression capabilities. The algorithm assembles in a top-down structure, initiating with the root node [12]. The root splits into sub-nodes, which repeatedly split again until the tree reaches a specified depth or until the model decides enough splitting has occurred to make viable leaves (predictions) [13].

An XGB algorithm, on the other hand, resembles an advanced form of gradient boosting: a procedure that repeatedly fits learning algorithms for enhanced accuracy [14]. In our case, XGB constantly creates decision trees and adjusts the weights of subsequent models depending on the performance of previous ones [15].

To determine the optimal training method for these models, we decided to train three default DT and XGB models in three varying circumstances: first on unmanipulated data, second on ROS data, and third on RUS data. Each time, the dataset was subject to an 80/20 split for training/testing respectively.

No Imbalanced Data Correction

The following visualizations illustrate the quantity of fraudulent and legitimate transactions provided in the training set that experienced no manipulation and the structure of the DT and XGB models created in the according circumstance.

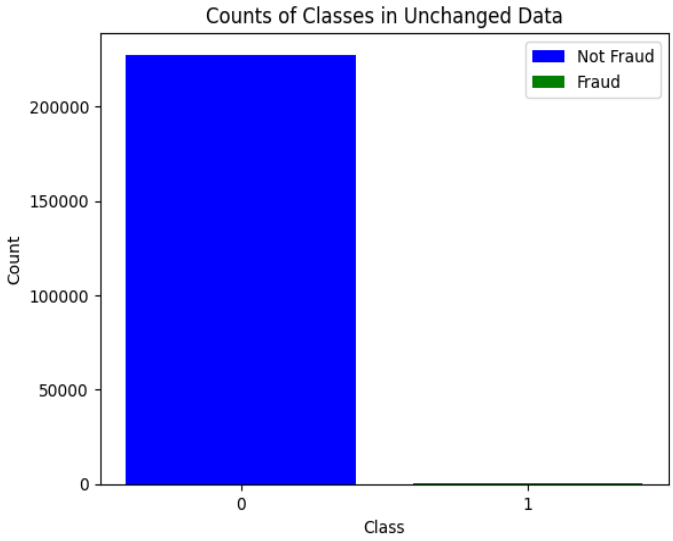


Figure 4. Training Set Class Distribution (no balancing)



Figure 5. DT Model Structure (no balancing)



Figure 6. XGB Model Structure (no balancing)

Random Over-Sampling

ROS treats an imbalanced dataset by randomly replicating minority class data until both classes are balanced [16].

The ensuing images display the number of transactions in each class after undergoing ROS and the structure of the DT and XGB models trained on it.

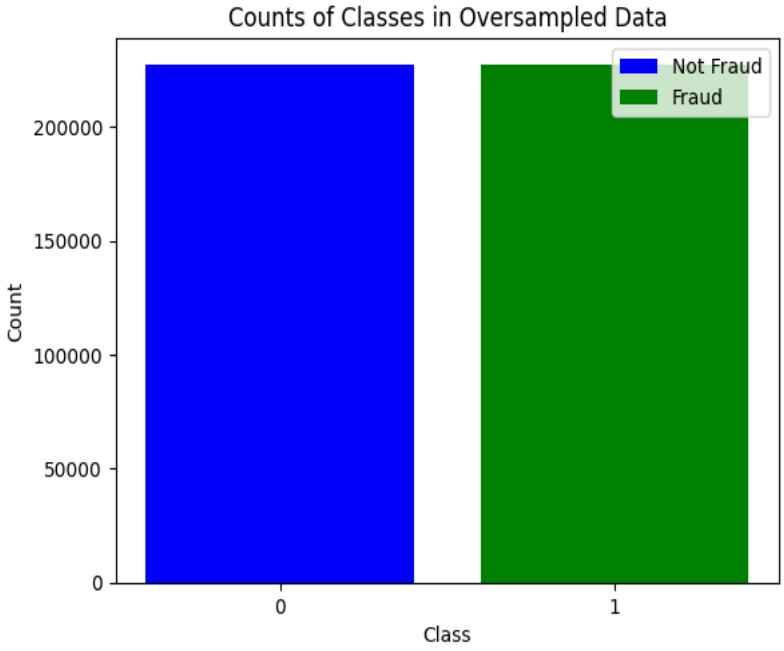


Figure 7. Training Set Class Distribution (ROS)

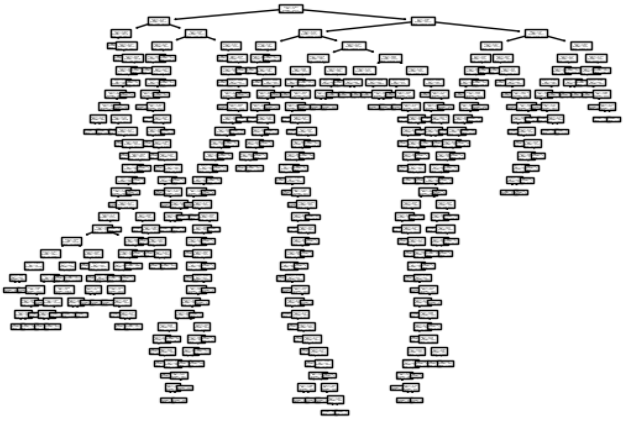


Figure 8. DT Model Structure (ROS)



Figure 9. XGB Model Structure (ROS)

Random Under-sampling

RUS is the rebalancing process of randomly selecting and removing data from the majority class until both classes are in equilibrium [17].

Figure 10 depicts the legitimate and fraudulent class quantities and figs 11 and 12 show the structures of DT and XGB models trained on RUS data.

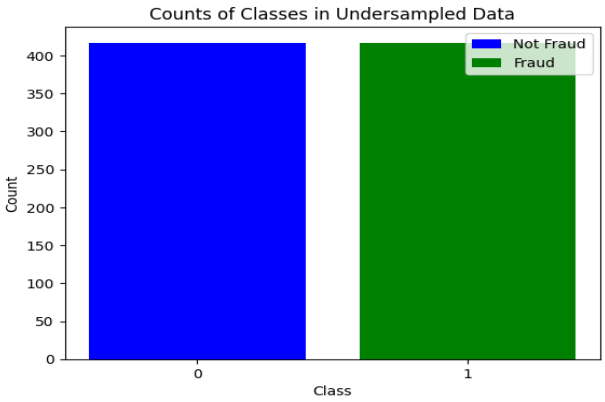


Figure 10. Training Set Class Distribution (RUS)

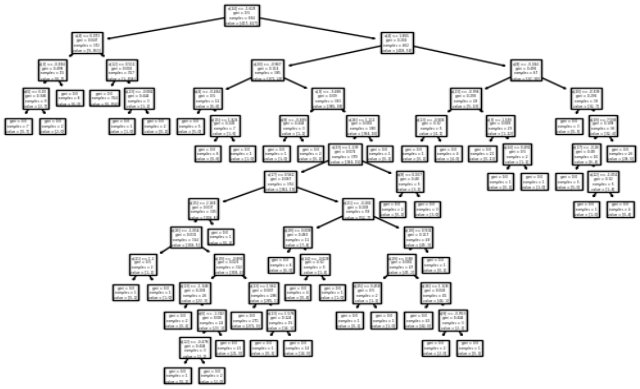


Figure 11. DT Model Structure (RUS)

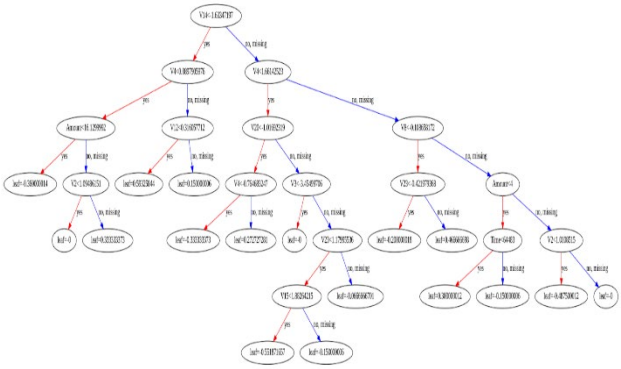


Figure 12. XGB Model Structure (RUS)

Results and Discussion

To diagnose the performance of the models we leveraged metrics such as accuracy, F1 score, precision, and recall. These are calculated using the components true positive (TP), false positive (FP), true negative (TN), and false negative (FN).

TP = Fraudulent transactions correctly predicted as fraudulent
FP = Legitimate transactions incorrectly predicted as fraudulent
TN = Legitimate transactions correctly predicted as legitimate
FN = Fraudulent transactions incorrectly predicted as legitimate

The first metric, accuracy, conveys the percentage of correct predictions out of the total number of predictions made [18].

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

This metric alone may be misleading due to the extreme class imbalance within the dataset. Accuracy is highly susceptible to bias produced by skewed data [19]. For instance, if the model was tested on the entire dataset, it could classify every transaction as legitimate and still receive an accuracy score of ~.9982.

Consequently, we incorporated other evaluation measurements like precision, recall, and F1 score to form a more holistic image of the models' abilities.

$$Precision = \frac{TP}{TP + FP}$$

$$Recall = \frac{TP}{TP + FN}$$

Our determination of the best models relied on a novel approach. We decided that the issue of fraud detection should prioritize high recall and forgive low precision to a greater extent than usual. High recall reflects a model's ability to effectively classify fraudulent transactions, minimizing FNs. Conversely, low precision suggests a high occurrence of FPs. While this may seem undesirable, it reinforces the principle of "better safe than sorry" by reducing questionable, possibly fraudulent, transactions from being mistakenly passed through as legitimate. Moreover, flagging transactions as potentially fraudulent augments the difficulty for fraudsters to escape with fraud. These instances provide cardholders with the opportunity to inspect the transactions and verify their true class. From a financial perspective, FPs may help mitigate losses associated with fraud marked as legit.

$$F1 = 2 \times \frac{precision \times recall}{precision + recall}$$

F1 Score combines both precision and recall, forming an overview of the model's capabilities in a single metric. This metric signifies the best model from a standard approach to model evaluation.

We also used data balancing methods ROS and RUS. These processes minimize the existence of peculiar behaviors within the machine learning algorithms [20]: overfitting and bias toward the majority class**.**

We observe that applying these techniques vastly alters the complexity of the ML models. Subjecting the models to ROS boosted the number of layers significantly in the regular DT as demonstrated by figs 5 and 8. However, despite the ROS XGB model looking increasingly complex, the depth of the model remained constant even after ROS as indicated by figs 6 and 9. In fact, the additional intricacy is likely explained by more splitting occurring and the generation of many more leaf nodes. Next, conducting RUS profoundly reduced the complexity of both models. Figures 5 and 11 illustrate the structures of unmanipulated and RUS DT models respectively. These depictions make it evident that carrying out RUS diminished the number of layers in the DT, opposite to the result of ROS. Figures 6 and 12 render the structures of unmanipulated and RUS XGB models respectively. It is visible that performing RUS abated the number of total nodes in the XGB model. Also, the number of layers in the RUS model is identical to those of the unmanipulated and ROS XGB models. The XGB algorithm containing a default max-depth as one of its parameters is probably responsible for this phenomenon.

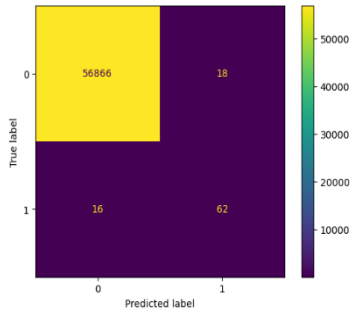


Figure 13. DT confusion matrix (no balancing)

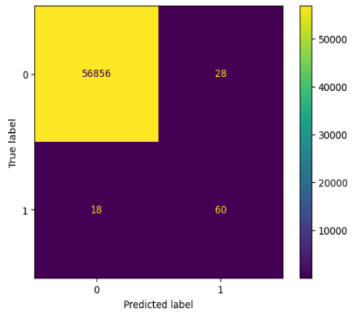


Figure 14. DT confusion matrix (ROS)

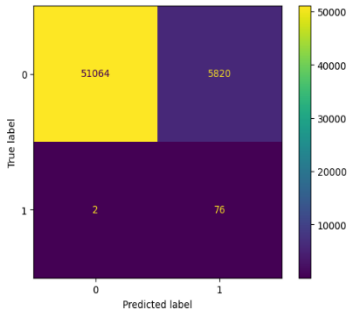


Figure 15. DT confusion matrix (RUS)

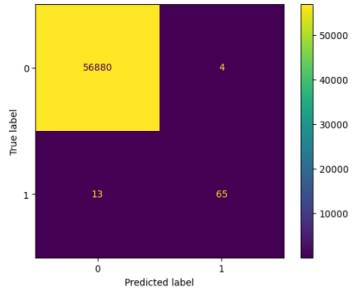


Figure 16. XGB confusion matrix (no balancing)

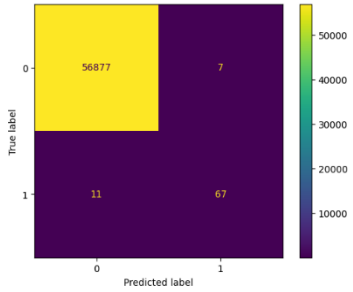


Figure 17. XGB confusion matrix (ROS)

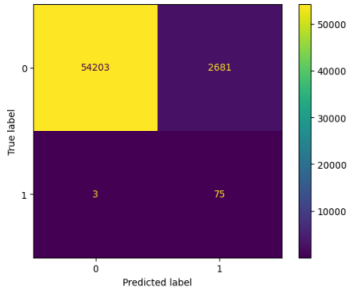


Figure 18. XGB confusion matrix (RUS)

Table 1. DT performance metrics in each circumstance

DecisionTree				
	Accuracy	Precision	F1 Score	Recall
No Balancing	0.99940	0.775	0.78481	0.79487
Random Over-sampling	0.99919	0.68181	0.72289	0.76923
Random Under-sampling	0.89844	0.01297	0.02560	0.97435

Table 2. XGB performance metrics in each circumstance

XGBoost				
	Accuracy	Precision	F1 Score	Recall
No Balancing	0.99970	0.94202	0.88435	0.83333
Random Over-sampling	0.99968	0.90540	0.88157	0.85897
Random Under-sampling	0.95288	0.02721	0.05292	0.96153

Based on our established evaluation approach and the results elicited by the confusion matrices, we discovered the top 3 models from best to worst to be RUS XGB, RUS DT, and ROS XGB. Tables 2 delineates that the best model, RUS XGB, received an accuracy score of 95.288%, a precision score of 2.721%, and F1 score of 5.292%, and a recall score of 96.1533%. Although RUS DT received a better recall score at 97.435%, we designated RUS XGB as the ultimate model because it properly classified 3,138 more transactions while having an essentially negligible difference in FNs.

From a standard approach, the top 3 models from best to worst are regular XGB, ROS XGB, and regular DT. Table 2 demonstrates that regular XGB achieved the greatest scores in each department among all the models with an accuracy of 99.97%, a precision of 94.202%, an F1 score of 88.435%, and a recall score of 83.33%.

Figures 13-18 above exhibit the XGB models performed better in nearly every category except for the under-sampled FN and TN sections in which the regular DT performed slightly better. The matrices also demonstrate that RUS had the greatest effect on the models' predictive outputs, particularly in the FP department. Figure 15 reports the RUS DT made over 5500 more FP predictions compared to its preceding models. Similarly, fig 18 reports the RUS XGB made over 2500 more FP predictions compared to its preceding models.

Limitations

Our decision to conduct ROS and RUS on the imbalanced data set and evaluate the machine learning algorithms using accuracy, precision, recall, F1-score, and confusion matrices permits numerous improvements. Future research endeavors could build on our findings by applying other imbalance correction methods such as Near-Miss and SMOTE. These create and remove data using processes different from ROS and RUS, thus, possibly fostering significantly different results. In addition, the employment of additional performance metrics like specificity, sensitivity, the receiver operating characteristic curve (ROC), area under the ROC, the precision-recall curve (PRC), and area under the PRC might supply further insight into the models' behavior. Finally, future research could utilize different algorithms altogether.

Conclusion

Employing Kaggle's Credit Card Detection dataset allowed us to leverage DT and XGB machine learning algorithms in an extremely imbalanced environment: ~.172% to ~99.82% for fraudulent to legitimate respectively. To eliminate bias established by skewed data we used ROS and RUS. Training the algorithms in various scenarios of dataset manipulation allowed us to discover that an XGB model trained on RUS data optimally detected fraud based on the notion that we prioritize high recall and forgive low precision. However, from a standard perspective, an XGB model trained on unmanipulated data performed the best.

These conclusions are rooted in performance metrics such as accuracy, precision, recall, F1-score, and confusion matrices. They enabled us to compare the different models and discover patterns facilitated by the balancing processes executed on the training sets. Whereas ROS promoted similar results to the unmanipulated circumstance, RUS resulted in a profound increase in FP predictions. The changes in the models' predictive nature raise concern for the true class of the transactions supporting the notion of forgiving low precision. Realistically, a cardholder would likely demonstrate gratitude for the opportunity to verify whether a potentially fraudulent transaction was truly fraudulent.

Finally, comparison of each model reveals that the XGB machine learning algorithm is superior to the DT algorithm.

Acknowledgments

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